

ST 308 Project

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Part 1

```
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.2
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
Warning: package 'tidyr' was built under R version 4.5.3
```

```
Warning: package 'readr' was built under R version 4.5.2
```

```
Warning: package 'dplyr' was built under R version 4.5.2
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Question 1

Quarto is a type of document in R that allows us to make documents or reports that can contain text, code, and show output all in one. Quarto document is useful for people who work with data because it saves the codes written, allowing people to go back and make any changes if needed. Quarto also helps us render to pdf file containing the graphical outputs of our codes, which is useful for making reports (This question is answered by Landrey Lord)

Question 2

```
library(readr)
house <- read_csv("aacline_house.csv")
```

Rows: 1000 Columns: 13

-- Column specification -----

Delimiter: ","

chr (6): HouseStyle, BsmtFinType2, Foundation, RoofStyle, Exterior1st, LandS...

dbl (7): SalePrice, BsmtFinSF1, GarageCars, YearBuilt, OpenPorchSF, WoodDeck...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
head(house)
```

A tibble: 6 x 13

	SalePrice	BsmtFinSF1	GarageCars	YearBuilt	OpenPorchSF	WoodDeckSF	YearRemodAdd
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	223500	486	2	2001	42	0	2002
2	140000	216	3	1915	35	0	1970
3	250000	655	3	2000	84	192	2000
4	200000	859	2	1973	204	235	1973
5	129500	906	1	1965	0	0	1965
6	345000	998	3	2005	21	147	2006

i 6 more variables: HouseStyle <chr>, BsmtFinType2 <chr>, Foundation <chr>,
RoofStyle <chr>, Exterior1st <chr>, LandSlope <chr>

(This question is answered by Hung Nguyen)

Question 3

GarageCars is a quantitative column, which has numerical values and specifically integer values in our dataset. HouseStyle is a categorical column, which has character values in our dataset (This question is answered by Hung Nguyen)

Question 4

```
house %>%
  filter(YearBuilt > median(YearBuilt),
         SalePrice < mean(SalePrice),
         HouseStyle == "2Story")
```

A tibble: 66 x 13

	SalePrice	BsmtFinSF1	GarageCars	YearBuilt	OpenPorchSF	WoodDeckSF	YearRemodAdd
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	172500	649	2	1999	0	115	2000
2	185000	0	2	1998	94	0	1998
3	174000	0	2	2005	38	100	2005
4	178000	0	2	1985	46	192	1985
5	176000	419	2	1999	32	0	1999
6	163990	0	2	2005	24	0	2006
7	130000	0	2	2004	40	0	2006
8	130000	0	2	2004	40	0	2004
9	148500	566	2	1976	0	87	1976
10	161750	378	1	1977	36	0	1977

i 56 more rows

i 6 more variables: HouseStyle <chr>, BsmtFinType2 <chr>, Foundation <chr>,

RoofStyle <chr>, Exterior1st <chr>, LandSlope <chr>

(This question is answered by Landrey Lord)

Question 5

```
house_subset <-
  house %>%
    rename(sale_price = SalePrice,
           garage_cars = GarageCars,
```

```

      year_built = YearBuilt) %>%
    mutate(garage_cars = factor(garage_cars)) %>%
    select(sale_price, garage_cars, year_built)
head(house_subset)

```

```

# A tibble: 6 x 3
  sale_price garage_cars year_built
      <dbl>   <fct>       <dbl>
1    223500 2          2001
2    140000 3          1915
3    250000 3          2000
4    200000 2          1973
5    129500 1          1965
6    345000 3          2005

```

(This question is answered by Hung Nguyen)

Question 6

```

tapply(X = house$WoodDeckSF, INDEX = list(house$Foundation, house$RoofStyle),
      FUN = mean, na.rm = TRUE)

```

	Gable	Hip	Other
BrkTil	58.51899	34.77778	29.4000
CBlock	85.46933	79.25253	110.2727
Other	51.05000	0.00000	90.5000
PConc	107.02825	155.95556	278.3333

```

tapply(X = house$WoodDeckSF, INDEX = list(house$Foundation, house$RoofStyle),
      FUN = sd, na.rm = TRUE)

```

	Gable	Hip	Other
BrkTil	118.47437	75.12952	65.7404
CBlock	138.98513	131.81274	141.1588
Other	92.71318	0.00000	127.9863
PConc	112.30901	124.87584	202.6976

(This question is answered by Hung Nguyen and Landrey Lord)

Part 2

Question 1

In the dataset, each observation has 7 attributes. Most variables are categorical, including START_DATE, END_DATE, CATEGORY, START, STOP, PURPOSE. Only MILES is a quantitative variable. In addition, START_DATE and END_DATE variables range from January to December 2016. Meanwhile, CATEGORY has two values: 'Business' and 'Personal'. START and STOP variables have values of places in the United States. MILES are quantitative values with many different travelled distances. PURPOSE contains specific trip reasons though many observations have missing values for this variable. The data came from a driver's personal Uber ride history. Every ride recorded by Uber includes: start time, end time, pickup location, drop-off location, miles traveled, trip purpose (if added manually). The research question is which are the top 3 most common travelling purposes, whether it is personal or business category, and which one has the longest average travelled distance. This research question relates to three variables: CATEGORY, PURPOSE, MILES. It answers which of the travelling purposes is popular and determines whether it is either a personal or business category. Audiences also can see which travelling purposes motivates people to travel longer distances by Uber, which costs them a considerable amount of money (This question is answered by Hung Nguyen and Landrey Lord)

Question 2

```
uber_data <- read_csv("UberDataset.csv")
```

```
Rows: 1156 Columns: 7
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (6): START_DATE, END_DATE, CATEGORY, START, STOP, PURPOSE
```

```
dbl (1): MILES
```

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
uber <- uber_data %>%  
  mutate(PURPOSE = na_if(PURPOSE, "")) %>%  
  # na_if() function is used to convert blank values in PURPOSE column into NA  
  # values to serve for drop_na() function and then perform statistical summary  
  drop_na(PURPOSE)  
  
table(uber$PURPOSE, uber$CATEGORY)
```

	Business	Personal
Airport/Travel	3	0
Between Offices	18	0
Charity (\$)	0	1
Commute	0	1
Customer Visit	101	0
Errand/Supplies	128	0
Meal/Entertain	160	0
Meeting	187	0
Moving	0	4
Temporary Site	50	0

```
uber %>%
  group_by(PURPOSE, CATEGORY) %>%
  summarise(miles_avg = mean(MILES))
```

`summarise()` has regrouped the output.
 i Summaries were computed grouped by PURPOSE and CATEGORY.
 i Output is grouped by PURPOSE.
 i Use `summarise(.groups = "drop\_last")` to silence this message.
 i Use `summarise(.by = c(PURPOSE, CATEGORY))` for per-operation grouping
 (`?dplyr::dplyr\_by`) instead.

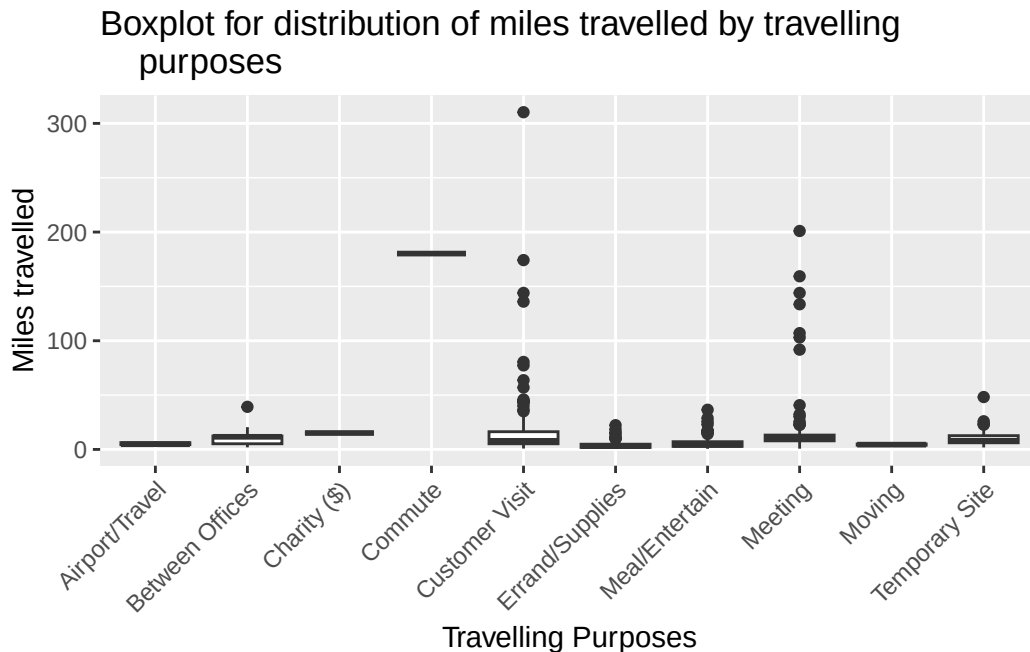
```
# A tibble: 10 x 3
# Groups:   PURPOSE [10]
  PURPOSE      CATEGORY miles_avg
  <chr>         <chr>      <dbl>
1 Airport/Travel Business      5.5
2 Between Offices Business     10.9
3 Charity ($)   Personal     15.1
4 Commute       Personal    180.
5 Customer Visit Business     20.7
6 Errand/Supplies Business      3.97
7 Meal/Entertain Business      5.70
8 Meeting       Business     15.2
9 Moving        Personal      4.55
10 Temporary Site Business     10.5
```

```
library(ggplot2)
ggplot(uber, aes(x = PURPOSE, y = MILES)) +
```

```

geom_boxplot() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1)) +
# we used this theme() function to format the x-axis of our boxplot to make it
# easier to see values on the x-axis
labs(
  title = "Boxplot for distribution of miles travelled by travelling
  purposes",
  x = "Travelling Purposes",
  y = "Miles travelled"
)

```



The most common travelling purpose is Meeting, the second one is Meal/Entertain, the third one is Errand/Supplies, all of these three purposes fall under the Business category. However, the highest average miles travelled is Commute which falls under the Personal category. The data presents that the top three most common ways of travelling are Meeting, Meal/Entertain, and Errand/Supplies, which belong to the Business category, none of them have the highest average miles travelled like we thought. The main reason behind this is that people do not travel long distances for these purposes, so the average miles travelled is not considerably high. With that people travel longer distances for Commute purposes, so the miles travelled is significantly high compared to the low frequency. With this specific case there is only 1 person travelling for Commute and that person travelled over 180 miles, making this the highest number we recorded of average miles travelled (This question is answered by Hung Nguyen and Landrey Lord)